

**TABLE 10.2 Summary of Piaget's Stages of Cognitive Development**

| STAGE AND APPROXIMATE AGE           | ACHIEVEMENTS AND ACTIVITIES                                                                                                              | LIMITATIONS                                                                                                                                                                                                               |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensorimotor (birth to 1½ years)    | Reacts to sensory stimuli through reflexes and other responses                                                                           | Little use of language; seems not to understand object permanence; does not distinguish appearance from reality                                                                                                           |
| Preoperational (1½ to 7 years)      | Develops language; can represent objects mentally by words and other symbols; can respond to objects that are remembered but not present | Lacks operations (reversible mental processes); lacks concept of conservation; focuses on one property at a time (such as length or width), not on both at once; still has trouble distinguishing appearance from reality |
| Concrete operations (7 to 11 years) | Understands conservation of mass, number, and volume; can reason logically with regard to concrete objects that can be seen or touched   | Has trouble reasoning about abstract concepts and hypothetical situations                                                                                                                                                 |
| Formal operations (11 years onward) | Can reason logically about abstract and hypothetical concepts; develops strategies; plans actions in advance                             | None beyond the occasional irrationalities of all human thought                                                                                                                                                           |

- b.** Child performs well on tests of object permanence; still has trouble with conservation.
- c.** Child has schemata; does not speak in complete sentences; fails tests of object permanence.
- d.** Child performs well on tests of object permanence, conservation, and hypothetical questions.

### Are Piaget's Stages Distinct?

According to Piaget, the four stages of intellectual development are distinct, and each transition from one stage to the next requires a major reorganization of the child's way of thinking. He contended that children in the sensorimotor stage fail certain tasks because they lack the concept of object permanence and that children in the preoperational stage fail conservation tasks because they lack the necessary mental processes. In other words, intellectual growth is marked by periods of revolutionary reorganization, as a child advances from one stage of thinking to another.

Later research has thrown much doubt upon this conclusion. If it were true, then a child in a given stage of development—say, the preoperational stage—should perform consistently at that level. In fact, a given child's performance will fluctuate, and one can increase that fluctuation by altering the difficulty of a task. For example, preschool children ordinarily fail conservation-of-number tasks, where an investigator presents two rows of coins or candies with seven or more objects in each row, then spreads out one row and asks which row “has more.” Preoperational children reply that the spread-out row has more. However, when Rochel Gelman (1982) presented two rows of only three objects each (Figure 10.19) and then spread out one of the rows, even 3- and

4-year-old children answered that the rows had the same number of items. (Most of the 3-year-olds counted first, to make sure.) After much practice with these short rows, most of the 3- and 4-year-olds also answered correctly that a spread-out row of eight items had the same number of items as the tightly packed row of eight.

In general, the progression from one stage of thinking to another appears to be gradual and not sudden (Siegler, 1994). That is, the difference between older children and younger children is not so much a matter of *having* an ability or of *lacking* it; the difference is between readily using the ability or using it only for simple tasks. As the children develop, they use the ability for increasingly more complex tasks.



**FIGURE 10.19** (a) With the standard conservation-of-number task, preoperational children answer that the lower row has more items. (b) With a simplified task, the same children say that both rows have the same number of items.

## Implications for Education: Piaget and Lev Vygotsky

One implication of Piaget's findings is that children must discover certain concepts, such as the concept of conservation, mainly on their own. Teaching any such concept means directing children's attention to the key aspects and then letting them discover the concept for themselves.

Another implication frequently drawn from Piaget's work is that teachers should determine a child's level of functioning and then teach material appropriate to that level. For example, teachers should not try introducing abstract concepts to children who are at the concrete operations stage of development.

In contrast to this view, the Russian psychologist Lev Vygotsky (1978) argued that education cannot simply wait for children to reach the next stage of development on their own; children have to learn in order to develop. Indeed, Vygotsky argued, the one distinguishing characteristic of human thought is that it is based on language and symbols. We use language both to influence others and to control our own behavior (as when we tell ourselves what to do and what not to do). Therefore, it is inefficient to expect children to explore and discover what we want them to know; adults should indeed teach them (Karpov & Haywood, 1998).

However, when Vygotsky said that adults should teach children, he did not necessarily mean to teach by lecturing them, and he certainly did not mean that adults should ignore the child's developmental level. What he meant was this: Every child has a **zone of proximal development**, which is the distance between what a child can do on his- or her own and what the child can do with the help of adults or other children. Instruction should always remain within that zone. For example, one should not try to teach a 4-year-old the concept of conservation of volume, and it would be a waste of time to try to teach it to an 8-year-old, who probably understands the concept already. But a child

around 6 years old who does not yet understand the concept might be able to learn it with help and guidance from an adult or older child. Similarly, children improve their recall of a story when adults provide appropriate hints and reminders, and they can solve more complicated math problems with help than they can solve alone. Vygotsky compared this help to *scaffolding*, the temporary supports that builders use when constructing a new building: After the building is complete, the scaffolding can be removed.

Good advice for an educator, therefore, is to be sensitive to a child's zone of proximal development, to try to detect how much further he or she can successfully push the child's development. In some cases, the child may provide clues. For example, several preoperational children who are asked which beaker has more water may all point to the tall, thin one; however, the children who could most easily learn conservation of volume describe the various beakers by using hand gestures that indicate both height and width (Goldin-Meadow, Alibali, & Church, 1993). An educator sensitive to those gestures could infer which children have the largest zone of proximal development and, therefore, which children are ready for certain kinds of instruction.

## The Development of Moral Reasoning

As children develop their reasoning powers, they apply their new reasoning abilities to moral issues. Just as 11-year-olds reason differently from 5-year-olds about what happens when water is poured from one beaker to another, they also reason differently about issues of right and wrong.

### Kohlberg's Method of Evaluating Levels of Moral Reasoning

Psychologists once regarded morality as a set of arbitrary, learned rules with no logical basis. Lawrence Kohlberg (1969; Kohlberg & Hersh, 1977) rejected that view, arguing instead that moral reasoning is the result of a reasoning process that resembles Piaget's stages of intellectual development. Young children mostly equate "wrong" with "punished." Adults understand that certain acts are wrong even though they may never lead to punishment and that other acts are right even if they do lead to punishment. Children younger than about 6 years old think that accidentally breaking something valuable is worse than intentionally breaking something of less value; older children and adults give more regard to people's intentions.

Kohlberg proposed that people pass through distinct stages as they develop moral reasoning. Although those stages are analogous to Piaget's stages, they do not follow the same time sequence. For example, an individual may progress rapidly through Piaget's stages but move more slowly through Kohlberg's stages.



Lev Vygotsky called attention to a child's zone of proximal development: The gap between what a child can do alone and what the child can do with help. Education within the zone of proximal development can advance a child's reasoning abilities.



Kohlberg suggested that moral reasoning should not be evaluated according to the decisions a person makes but according to the reasoning behind them. For example, do you think that designing nuclear bombs is a moral way to make a living? According to Kohlberg, whether it shows good moral reasoning depends on one's reasons for taking the job. Hugh Gusterson (1992) interviewed nuclear bomb designers at the Lawrence Livermore National Laboratory. When he asked them about the morality of their actions, nearly everyone said something like, "You're lucky you chose me to interview, because I, unlike all the others, think deeply about these matters." (Clearly, all of them were thinking about the morality of their profession, but they were not discussing it with one another.) The vast majority said they were confident that the weapons they designed would never be used and that the only function of the bombs was to threaten and, thereby, to prevent a war. If you disagree with these bomb designers' assumptions, you might disagree with the morality of their *actions*, but according to Kohlberg's system, you should concede that they are operating at a high level of moral *reasoning*, because they explained their actions in terms of the expected benefits to humanity.

Kohlberg believed that we all start with a low level of moral reasoning and mature through higher stages. To measure the maturity of a person's moral judgments, Kohlberg devised a series of **moral dilemmas**—problems that pit one moral value against another. Each dilemma is accompanied by a question, such as "What should this person do?" or "Did this person do the right thing?" Kohlberg was not concerned about the choice a person makes, because the dilemmas pit one value against another and well-meaning people do disagree on the right answer. More revealing than the answer itself is the explanation behind it. The respondent's explanations are then matched to one of Kohlberg's six stages, which are grouped into three levels (see Table 10.3). Because very few people operate consistently at stage 6, many authorities combine stages 5 and 6. To emphasize: In Kohlberg's scheme, there are no moral or immoral decisions, just moral and immoral *reasons* for making a decision. (See Figure 10.20.)

### SOMETHING TO THINK ABOUT

Suppose a military junta overthrows a democratic government and sets up a dictatorship. In which of Kohlberg's stages of moral reasoning would you classify the members of the junta? Would your answer depend on the reasons they gave for setting up their dictatorship? ★

People's responses to Kohlberg's moral dilemmas suggest the level of moral reasoning at which they *usually* operate. Few people are absolutely consistent in their moral reasoning, any more than they are consistent in any other kind of reasoning.

Kohlberg and others have found that people begin at



**FIGURE 10.20** Here is a real-life moral dilemma: Michigan physician Jack Kevorkian developed a device to help terminally ill patients kill themselves painlessly. Is it morally right or wrong to aid in someone else's suicide? According to Kohlberg's viewpoint, the morality of an act does not depend on the act itself, but on the reasoning behind the act.

the first stage and progress through the others in order, although they do not all reach the highest stages. (The order of progression is an important point: If people were just as likely to progress in the order 3-5-4 as in the order 3-4-5, then we would have no justification for regarding stage 5 as more advanced than stage 4.) People seldom skip a stage or revert to an earlier stage after reaching a higher one.

Figure 10.21 shows that most 10-year-olds' judgments are at Kohlberg's first or second stage, but that the mode of 16-year-olds' judgments is at stage 5. What accounts for this rather swift development of moral reasoning? Kohlberg suggests that it results from cognitive growth: 16-year-olds are capable of more mature reasoning than 10-year-olds. Rachael Henry (1983) proposed a different possibility: Adolescents reject parental authority. In stages 1 and 2, parents are the source of moral judgments, and what they say determines what is right and what is wrong. In stage 3, other people become the source of moral judgments. In stage 4, the source is the law; in stages 5 and 6, society as a whole or some abstract truth is the source of moral judgments. As adolescents continue to mature, they move farther and farther away from regarding their parents as the ultimate authority on questions of morality.

### CONCEPT CHECK

- For the moral dilemma described at the top of Table 10.3, suppose someone says that Heinz was wrong to steal the drug to save his wife's life. Which level of moral reasoning is characteristic of this judgment? (Check your answer on page 367.)

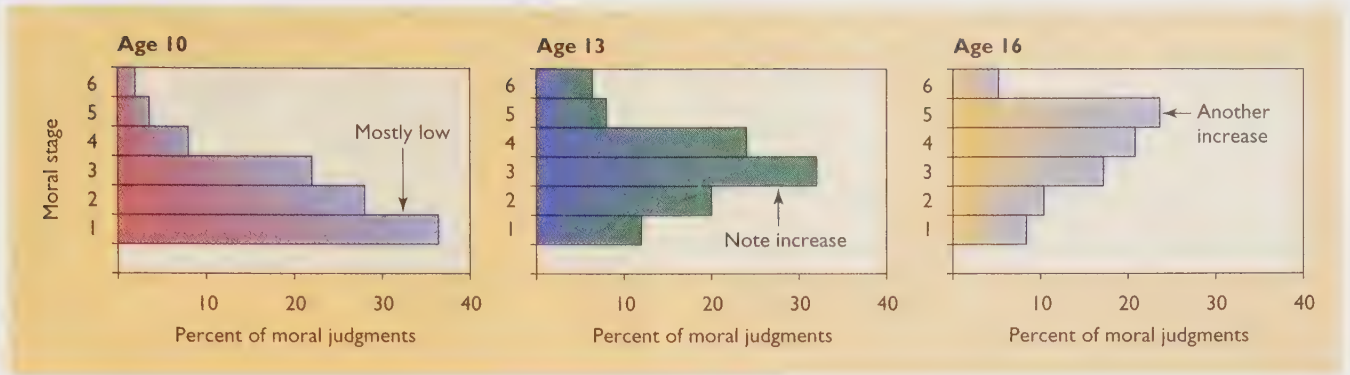
**TABLE 10.3 Responses to One of Kohlberg's Moral Dilemmas by People at Six Levels of Moral Reasoning**

*The dilemma:* Heinz's wife was near death from cancer. A druggist had recently discovered a drug that might be able to save her. The druggist was charging \$2,000 for the drug, which cost him \$200 to make. Heinz could not afford to pay for it, and he could borrow only \$1,000 from friends. He offered to pay the rest later. The druggist refused to sell the drug for less than the full price paid in advance: "I discovered the drug, and I'm going to make money from it." Late that night, Heinz broke into the store to steal the drug for his wife. Did Heinz do the right thing?

| LEVEL/STAGE                                                       | TYPICAL ANSWER                                                                                                                                          | BASIS FOR JUDGING RIGHT FROM WRONG                                                                                                             | DESCRIPTION OF STAGE                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The Level of Preconventional Morality</b>                      |                                                                                                                                                         |                                                                                                                                                |                                                                                                                                                                                                                                                                               |
| 1. Punishment and obedience orientation                           | "No. If he steals the drug he might go to jail."<br>"Yes. If he can't afford the drug, he can't afford a funeral, either."                              | Wrong is equated with punishment.<br>What is good is whatever is in the man's immediate self-interest.                                         | Decisions are based on their immediate consequences.<br>Whatever is rewarded is "good" and whatever is punished is "bad." If you break something and are punished, then what you did was bad.                                                                                 |
| 2. Instrumental relativist orientation                            | "He can steal the drug and save his wife, and he'll be with her when he gets out of jail."                                                              | Again, what is good is whatever is in the man's own best interests, but his interests include delayed benefits.                                | It is good to help other people, but only because they may one day return the favor: "You scratch my back and I'll scratch yours."                                                                                                                                            |
| <b>The Level of Conventional Morality</b>                         |                                                                                                                                                         |                                                                                                                                                |                                                                                                                                                                                                                                                                               |
| 3. Interpersonal concordance, or "good boy/nice girl" orientation | "People will understand if you steal the drug to save your wife, but they'll think you're cruel and a coward if you don't."                             | Public opinion is the main basis for judging what is good.                                                                                     | The "right" thing to do is whatever pleases others, especially those in authority. Be a good person so others will think you are good. Conformity to the dictates of public opinion is important.                                                                             |
| 4. "Law and order" orientation                                    | "No, because stealing is illegal."<br>"Yes. It is the husband's duty to save his wife even if he feels guilty afterward for stealing the drug."         | Right and wrong can be determined by duty or by one's role in society.                                                                         | You should respect the law — simply because it is the law — and work to strengthen the social order that enforces it.                                                                                                                                                         |
| <b>The Level of Postconventional or Principled Morality</b>       |                                                                                                                                                         |                                                                                                                                                |                                                                                                                                                                                                                                                                               |
| 5. Social-contract legalistic orientation                         | "The husband has a right to the drug even if he can't pay now. If the druggist won't charge it, the government should look after it."                   | Laws are made for people's benefit. They should be flexible. If necessary, we may have to change certain laws or allow for exceptions to them. | The "right" thing to do is whatever people have agreed is the best thing for society. As in stage 4, you respect the law, but in addition recognize that a majority of the people can agree to change the rules. Anyone who makes a promise is obligated to keep the promise. |
| 6. Universal ethical principle orientation                        | "Although it is legally wrong to steal, the husband would be morally wrong not to steal to save his wife. A life is more precious than financial gain." | Right and wrong are based on absolute values such as human life. Sometimes these values take precedence over human laws.                       | In special cases, it may be right to violate a law that conflicts with higher ethical principles, such as justice and respect for human life. Among those who have obeyed a "higher law" are Jesus, Mahatma Gandhi, and Martin Luther King, Jr.                               |

SOURCE: Kohlberg, 1981





**FIGURE 10.21** Distinguishing right from wrong: The development of moral reasoning. Most younger adolescents give answers corresponding to Kohlberg's earlier moral stages. By age 16, most are at Kohlberg's fourth and fifth stages. (Based on Kohlberg, 1969.)

### The Limitations of Kohlberg's Views of Moral Development

Kohlberg's theories have had an enormous impact on psychology, although they do have certain limitations. Some critics point out that moral reasoning is just one part of moral behavior. James Rest (1983) divides moral behavior into four components:

1. Interpreting the situation
2. Deciding on the morally correct thing to do
3. Deciding what you will actually do, which may not be the same as the morally correct thing to do
4. Actually doing what you have decided to do

Kohlberg's stages relate only to the first and second of these components. Many juvenile delinquents and adult criminals make mature responses to Kohlberg's moral dilemmas but then engage in destructive behavior (Jurkovic, 1980; Link, Sherer, & Byrne, 1977). Apparently, a person can distinguish between right and wrong in the abstract and then behave in a way that ignores that reasoning (Figure 10.22).

Another criticism is that Kohlberg may have misinterpreted the lower stages of moral reasoning because they are characteristic of children or others who have trouble articulating their views. For example, many children say that stealing is wrong "because you might get caught." However, if children are asked whether they would steal something if they were sure they would not be caught, many say no. Apparently, they have a feeling of moral obligation, a sense of right and wrong, even if they are unable to state this idea clearly. In many cultures, such as the Hindu culture of India, almost all moral reasoning is stated in terms of natural obligations, not in terms of the laws and social conventions that figure so heavily in Western thought. For example, in Hindu custom, people should never eat beef, widows should not eat fish, and menstruating women should seclude themselves and not touch the family's meals. The Hindus defend these customs, which seem arbitrary to outsiders, as part of a natural moral order. In short, contrary to

Kohlberg's views, the idea of moral obligation may be a natural concept of human thought, even in childhood—and not something that we develop later in life after passing through the stage of conventional reasoning (Shweder, Mahapatra, & Miller, 1987).

Yet another criticism is that Kohlberg concentrated entirely on one type of moral reasoning, which we might call the "justice" orientation, based on people's rights. Carol Gilligan (1977, 1979) pointed out a different way of approaching moral decisions, the "caring" orientation: what would help or hurt other people. For example, consider a situation during the Vietnam War, when a group of soldiers were ordered to kill a group of unarmed civilians. One soldier, who regarded the order as immoral, refused to shoot. So, in terms of "justice," this soldier acted at a high moral level, following a "higher law" that required him not to kill. However, his actions made no difference, as the other soldiers killed all the civilians. In terms of "caring," he would have been more moral if he had found a way to hide a few of the Vietnamese civilians (Linn & Gilligan, 1990).



**FIGURE 10.22** Critics of Kohlberg's approach to moral reasoning point out that moral reasoning is not the same thing as moral action. This cadet at a Russian military academy could probably explain why cheating is immoral, but he cheats nevertheless.

Initially, Gilligan (1977, 1979) proposed that the “justice” and “caring” orientations represented a sex difference. Men, she said, focus mostly on rights and duties; women focus more on caring and relationships. For example, when asked about the ethics of abortion, men might say that abortion is wrong because it takes a life, or else they might say it is acceptable because a woman has the right to make decisions about her own body. Either answer would get a high evaluation in Kohlberg’s system, if explained clearly. But a woman might answer the same question by saying that the ethics of abortion depends on many of the details of each situation. An “it depends” answer does not get high marks in Kohlberg’s system, because his system does not rely on abstract principles of right and wrong. Nevertheless, this type of answer may reflect a sympathetic, caring approach to solving people’s problems.

Gilligan therefore proposed alternative stages of moral development, outlined in Table 10.4. The postconventional stage is the most mature; the preconventional stage is the least mature. Like Kohlberg, Gilligan concentrates on the reasons behind someone’s moral decisions, not on the decisions themselves. But unlike Kohlberg, Gilligan emphasizes the “caring” aspect of the reasons: Will this action help or hurt the people it affects?

Later researchers have agreed with Gilligan that people have two ways of reasoning about moral issues, one based on justice and the other based on caring, although they have failed to confirm any consistent difference between men’s and women’s moral reasoning (Shweder & Haidt, 1993). Nearly everyone, male or female, shows concern with both justice and caring.

Gilligan has therefore modified her position: Each of us has within ourselves two “voices” of morality—a voice of justice and rights and a voice of caring and relationships. One voice may speak a little louder in some people, but both voices are valid (Gilligan & Attanucci, 1988). Sometimes, the two voices are in conflict; that is, one may feel an obligation to help a person and yet find that the only way to help requires breaking a general principle of justice (Linn & Gilligan, 1990). In short, moral decisions can be complicated. (See Figure 10.23.)



**FIGURE 10.23** Sometimes the two “voices” of moral reasoning—justice and caring—are in conflict with each other. From a caring standpoint, you want to help a person in distress. From a justice standpoint, though, you may think it wrong to encourage begging. In many controversial situations, ranging from abortion to the death penalty to animal protection, well-meaning people disagree about what actions are moral.

### THE MESSAGE

#### *Developing Cognitive Abilities*

The universe is a complicated place, and even well-educated adults continue working throughout their lives to understand it. For a young child, this work is an enormous challenge, and we adults find it difficult to reconstruct what the world looks like through a child’s eyes. In childhood, as throughout life, a person is constantly constructing one hypothesis after another, discarding old conceptions of the world to make way for new ones.

### S U M M A R Y

- ★ *Piaget’s view of children’s thinking.* According to Jean Piaget, children’s thought processes are more intuitive, less logical, and more egocentric than adults’ thought processes are. (page 354)
- ★ *Piaget’s stages of development.* Piaget described four stages of development of thought processes and knowledge. Children progress through those stages in order. (page 355)
- ★ *Egocentric thinking.* Young children sometimes have trouble understanding other people’s point of view. Before about age 4, children have trouble inferring what someone else is likely or unlikely to know. (page 356)
- ★ *Restricted uses of abilities.* Young children do not always use all the abilities they have. Although children in the preoperational stage ordinarily fail to demonstrate their understanding of conservation of number, it is clear from other tasks that they do understand the concept of number. (page 361)

**TABLE 10.4** Carol Gilligan’s Stages of Moral Development

| STAGE            | BASIS FOR DECIDING RIGHT FROM WRONG              |
|------------------|--------------------------------------------------|
| Preconventional  | What is helpful or harmful to myself?            |
| Conventional     | What is helpful or harmful to other people?      |
| Postconventional | What is helpful or harmful to myself and others? |



★ **Implications for education.** Teaching must be at a level appropriate to the developmental stage of the child. However, as Lev Vygotsky argued, children can advance with an adult's help and thus do more than they could on their own. Therefore, educators should try to push children somewhat beyond their current level of performance. (page 362)

★ **Kohlberg's view of moral reasoning.** Lawrence Kohlberg contended that moral reasoning can also be described in terms of stages. According to Kohlberg, a person's moral reasoning should be evaluated on the basis of the reasons the person gives for a decision, rather than on the basis of the decision itself. (page 362)

★ **Gilligan's view of moral reasoning.** Carol Gilligan demonstrated that not all people decide moral dilemmas primarily on the basis of principles of justice. Some decide primarily on the basis of a caring orientation. (page 365)

## Suggestion for Further Reading

Hobson, R. P. (1993). *Autism and the development of mind*. Hove, East Sussex, UK: Lawrence Erlbaum Associates. Despite the title, this book is only partly about autism; it is about how children come to understand other people and to communicate with them.

## Terms

**schema** (plural: **schemata**) an organized way of interacting with objects in the world (page 354)

**assimilation** Piaget's term for the application of an established schema to new objects (page 355)

**accommodation** Piaget's term for the modification of an established schema to fit a new object (page 355)

**sensorimotor stage** according to Piaget, the first stage of intellectual development; an infant's behavior is limited to making simple motor responses to sensory stimuli (page 355)

**preoperational stage** according to Piaget, the second stage of intellectual development, in which children lack operations (page 356)

**operation** according to Piaget, a mental process that can be reversed (page 356)

**egocentric** the inability to take the perspective of another person, a tendency to view the world as centered around oneself (page 356)

**conservation** the concept that objects retain their weight, volume, and certain other properties in spite of changes in their shape or arrangement (page 358)

**stage of concrete operations** according to Piaget, the third stage of intellectual development; children can deal with the properties of concrete objects but cannot readily comprehend hypothetical or abstract questions (page 360)

**stage of formal operations** according to Piaget, the fourth and final stage of intellectual development; in this stage, people deal with abstract, hypothetical situations, which demand logical, deductive reasoning and systematic planning (page 360)

**zone of proximal development** the distance between what a child can do on his or her own and what the child can do with the help of adults or older children (page 362)

**moral dilemma** a problem that pits one moral value against another (page 363)

## Answers to Concept Checks

- (c) is the clearest case of egocentric thought, a failure to recognize another person's point of view. It is not the same as selfishness. (page 357)
- a.** Stage of concrete operations; **b.** preoperational stage; **c.** sensorimotor stage; **d.** stage of formal operations. (page 360)
- Not enough information is provided to answer this question. In Kohlberg's system, any judgment can represent either a high or a low level of moral reasoning; we evaluate a person's moral reasoning entirely by the explanation for the judgment, not by the judgment itself. (page 363)

## Web Resources

### Jean Piaget's Theory of Development

[snycorva.cortland.edu/~ANDERSMD/PIAGET/OPEN.HTML](http://snycorva.cortland.edu/~ANDERSMD/PIAGET/OPEN.HTML)  
A student-created, concise tutorial on the basic concepts advanced by Piaget covers the stages of intellectual development, conservation experiments, and a biological theory of intelligence.

## Social and Emotional Development

What special social and emotional problems do people face at different stages of their life?

What determines how we develop socially and emotionally?

**Y**ou are a contestant on a new TV game show called “What’s My Worry?” Behind the curtain is someone with an overriding concern. You are to identify that concern by questioning a psychologist who knows what it is. (You can neither see nor hear the person.) You must ask questions that can be answered with a single word or a short phrase. If you identify the worry correctly, you can win as much as \$50,000.

But there is one catch: The more questions you ask, the smaller your prize will be. If you guess correctly after only one question, you win \$50,000. After two questions, you win \$25,000. And so on. It would therefore be poor strategy to keep asking questions until you were sure of the



As people grow older, they mature in their social and emotional behaviors. However, many revert quickly to childlike behaviors in situations where such behavior is acceptable.

answer; instead, you should ask one or two questions and then guess.

What would your first question be? Mine would be: “How old is this person?” The principal worries of teenagers are different from those of most 20-year-olds, which in turn differ from those of most 40-year-olds and 70-year-olds. Each age has its own characteristic concerns, opportunities, and pleasures.

## Research Designs for Studying Development

Comparing the psychology of people of different ages sounds easy: We study a group of, say, 10-year-olds and a group of 20-year-olds to see how they differ. But how do we know whether those differences are due to age or to another difference between the groups? We may get different results depending on exactly how we conduct the study. Depending on the circumstances, psychologists can use either cross-sectional studies or longitudinal studies. A **cross-sectional study compares groups of individuals of different ages all at the same time**. For example, we could compare the drawing abilities of 6-year-olds, 8-year-olds, and 10-year-olds. The main weakness of cross-sectional studies is the difficulty of obtaining equivalent samples at different ages. For example, suppose you want to do a cross-sectional study comparing 20-year-olds and 60-year-olds. You study a sample of 20-year-olds from the local college, but how will you find a comparable group of 60-year-olds? The 60-year-olds you eventually select may have been less educated from the start, perhaps less wealthy, perhaps different in many other regards.

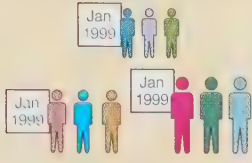
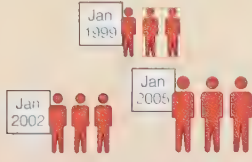
A **longitudinal study follows a single group of individuals as they develop**. For example, we could study a group of 6-year-olds and then study the same children again when they reach ages 9 and 12. Table 10.5 contrasts the two kinds of studies.

Longitudinal studies also face certain practical difficulties. A longitudinal study of children from the ages of 6 to 12 necessarily requires 6 years to complete; a longitudinal study from 6 to 60 would take much too long for any one investigator to complete. To make matters worse, many of the children who begin in a study at, say, age 6 may move out of town by age 9 or 12. A longitudinal study of the elderly faces the likelihood that many people will die or become seriously ill before the end of the study. Furthermore, those who leave a study may be different in important ways from those who continue. For example, suppose a creature from outer space observes humans for the first time and discovers that about 50% of young adults are males but that only 10–20% of 90-year-olds are males. The creature then concludes that, as human males grow older, most of them transform into females.

You can see why that conclusion is wrong. Males—with a few exceptions—do not change into females, but on the av-



**TABLE 10.5 Cross-Sectional and Longitudinal Studies**

|                        | DESCRIPTION                                                                                                                                             | ADVANTAGES                                                                                                                                                                                                               | DISADVANTAGES                                                                                                                                                                       | EXAMPLE                                                                                                    |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Cross-sectional</b> |  <p>Several groups of subjects of various ages studied at one time</p> | <ol style="list-style-type: none"> <li>1. Quick</li> <li>2. No risk of confusing age effects with effects of changes in society</li> </ol>                                                                               | <ol style="list-style-type: none"> <li>1. Risk of sampling error by getting different kinds of people at different ages</li> <li>2. Risk of cohort effects</li> </ol>               | Compare memory abilities of 3-, 5-, and 7-year-olds                                                        |
| <b>Longitudinal</b>    |  <p>One group of subjects studied repeatedly as they grow older</p>    | <ol style="list-style-type: none"> <li>1. No risk of sampling differences</li> <li>2. Can study effects of one experience on later development</li> <li>3. Can study consistency within individuals over time</li> </ol> | <ol style="list-style-type: none"> <li>1. Takes a long time</li> <li>2. Some subjects quit</li> <li>3. Sometimes hard to separate effects of age from changes in society</li> </ol> | Study social and emotional behavior of children at time of parents' divorce and at various times afterward |

erage they die earlier, leaving a greater percentage of older females. Similarly, in any longitudinal study, it is possible that those who left a study (for whatever reason) are brighter or less bright, more motivated or less motivated, or different in a variety of other ways from those who continued in the study. If so, the differences between the average person at the beginning of the study and the average person at the end of the study are not really due to the process of aging. **Selective attrition**, also known as *differential survival*, is the *tendency for some kinds of people to be more likely than others to drop out of a study*. Psychologists can partly reduce this problem by reporting the data only for the people who completed the study and disregarding the data for those who left.

A longitudinal study also faces the *difficulty of separating the effects of age from the effects of changes in society*. For example, suppose we found that a group of people who were 20 years old in 1970 became politically more conservative by age 40. We would not know whether these people became more conservative because of their age or because of changes in the political situation between 1970 and 1990.

Why, then, would investigators ever conduct a longitudinal study? One reason is that certain questions logically require a longitudinal study. For example, to study the effects of divorce on children, we can learn much by comparing how each child reacts at first with how that same child reacts several years later. To study whether happy children are likely to become happy adults, we would have to follow a single group of people for a substantial period of time.

What happens if a cross-sectional study gets one result and a longitudinal study gets another? Psychologists look

for the reasons behind the different results. For example, suppose someone conducting a study in 1955 finds that the mean IQ score of the 20-year-olds is much higher than that of the 60-year-olds. However, on a retest in 1995, the former 20-year-olds (now 60 years old) score as high as they did in 1955. Thus, the longitudinal part of the study indicates that IQ scores are fairly stable from age 20 to age 60. Why, then, did the 20-year-olds score so much higher than the 60-year-olds in 1955? There are two possible reasons: First, perhaps in 1955 the investigators selected a particularly bright group of 20-year-olds or a dull group of 60-year-olds. Secondly, maybe the generation of people who were 20 years old in 1955 were healthier and better educated than the older generation had ever been. That is, the 20-year-olds of 1955 may have performed better than the 20-year-olds of 1915 would have. Psychologists call this a *cohort effect* (Figure 10.24). **A cohort is a group of people born at a particular time or a group of people who entered an organization at a particular time.** Psychologists try to distinguish whether a difference among people of different ages is really due to age or whether it is a difference among cohorts.

## CONCEPT CHECKS

5. Suppose you want to study the effect of age on artistic abilities, and you want to be sure that any apparent differences are due to age and not to a cohort effect. Which should you use, a longitudinal study or a cross-sectional study?



**FIGURE 10.24** Children of the 1920s, 1960s, and 1990s differed in their behavior because they grew up in different historical eras, with different options regarding education, nutrition, and health care. Differences among age groups based on such influences are called cohort effects.

6. Suppose you want to study the effect of age on choice of clothing, but you are concerned that clothing fashions change from one year to the next. (It would not be fair to compare the people of today with people of previous years.) Which should you use, a longitudinal study or a cross-sectional study?
7. At Santa Enigma College, the average first-year student has a C-minus average, and the average senior has a B-plus average. An observer concludes that, as students progress through college, they improve their study habits. Based on the idea of selective attrition, propose another possible explanation. (Check your answers on page 381.)

Using a combination of cross-sectional and longitudinal studies, psychologists study changes in people's abilities and also changes in their life situations. The decisions and social environment that young adults face are very different from those experienced by children on the one hand or elderly people on the other hand.

## Erikson's Ages of Human Development

How people spend their time is largely determined by their current role in life—preschool child, student, worker, or retired person. Moreover, a person's role in life is determined largely, though not entirely, by his or her age. To understand why people behave as they do, we need to know the decisions they are facing at the current stage of their life.

Erik Erikson (Figure 10.25) divided the human life span into eight ages, each with its own social and emotional conflicts. First is the age of the newborn infant, whose main conflict is basic trust versus mistrust. The infant asks, in effect, "Is my social world predictable and supportive?" An infant whose early environment is supportive, nurturing, and loving will form a strong attachment to the parents that will also positively influence future relationships with other people (Erikson, 1963).

Erikson's second age is the age of the toddler, 1–3

years old, whose main conflict is autonomy versus shame and doubt. The toddler faces the issue, "Can I do things by myself or must I always rely on others?" Experiencing independence for the first time, the toddler begins to walk and talk, to become toilet trained, to obey some instructions and defy others. Depending on how the parents react, children either develop a healthy feeling of autonomy (independence) or a self-critical sense of shame and the doubt that they can accomplish things on their own.

Erikson's third age is the age of the preschool child, whose main conflict is initiative versus guilt. At ages 3–6, as children begin to broaden their horizons, their boundless energy comes into conflict with parental restrictions. Sooner or later, the child breaks something or makes a big mess. The child then faces the question, "Am I good or bad?" In contrast to the previous stage, where the child was concerned about what he or she is capable of doing, at this stage the child is concerned about the morality or acceptability of his or her actions.

In the fourth age, preadolescence (about ages 6–12), industry versus inferiority is the main conflict. The



**FIGURE 10.25** Erik Erikson argued that each age group has its own special social and emotional conflicts.



question is: “*Am I successful or worthless?*” Children widen their focus from the immediate family to society at large and begin to prepare for adult roles. They fantasize about the great successes ahead, and they begin to compete with their peers in an effort to excel in the activities popular with their age group. Children begin to develop a sense of either competence or inadequacy.

Erikson’s fifth age is adolescence (the early teens), where the main conflict is identity versus role confusion. Adolescents begin to seek independence from their parents and try to answer the question “*Who am I?*” or “*Who will I be?*” They may eventually settle on a satisfactory answer—form an identity—or they may continue to experiment with goals and lifestyles—a state of role confusion.

Erikson’s sixth age is young adulthood, in which intimacy versus isolation is the main conflict: “*Shall I share my life with another person or live alone?*” Young adults who marry or live with a friend find that they must adjust their habits to make the relationship succeed. Those who choose to live alone may experience loneliness and pressure from their parents and friends to find a suitable partner.

Erikson’s seventh age is middle adulthood. Here, the major conflict is generativity versus stagnation: “*Will I produce something of real value? Will I succeed in my life, both as a parent and as a worker? Will I in some way give something back to society?*”

Finally comes old age (the years after retirement), where the main conflict is ego integrity versus despair. “*Have I lived a full life or have I failed?*” Integrity is a state of contentment about one’s life, past, present, and future. Despair is a state of great disappointment about the past and the present, coupled with fear of the future. Table 10.6 summarizes Erikson’s ages.

Is Erikson’s view of development accurate? This question is unanswerable. Some psychologists find Erikson’s de-

scription of development a useful way to organize our thinking about human life; others find it less useful, and it is not the kind of theory that one can test scientifically. Erikson described development; he did not explain it. For example, he hardly addressed the question of how or why a person progresses from one stage to the next.

Now let’s take a closer look at some of the major issues that confront people in their social and emotional development at different ages. Beyond the primary conflicts that Erikson highlighted, development is marked by a succession of other significant problems.

### SOMETHING TO THINK ABOUT

Suppose you disagreed with Erikson’s analysis; for example, suppose you believe that the main concern of young adults is not “intimacy versus isolation” but “earning money versus not earning money” or “finding meaning in life versus meaninglessness.” How might you determine whether your theory or Erikson’s is more accurate? ★

## Infancy: Forming the First Attachments

Before the late 1950s, if someone had asked, “What causes an infant to develop an attachment to its mother?” almost all psychologists would have replied, “Mother’s milk.” They were wrong.

### Studies of Attachment Among Monkeys

Attachment—a long-term feeling of closeness between people, such as a child and a caregiver—depends on more than just being fed. Attachment is part of trust, in Erikson’s

**TABLE 10.6 Erikson’s Ages of Human Development**

| AGE                                    | MAIN CONFLICT                   | TYPICAL QUESTION                                             |
|----------------------------------------|---------------------------------|--------------------------------------------------------------|
| Infant                                 | Basic trust versus mistrust     | Is my social world predictable and supportive?               |
| Toddler (ages 1–3)                     | Autonomy versus shame and doubt | Can I do things by myself or must I always rely on others?   |
| Preschool child (ages 3–6)             | Initiative versus guilt         | Am I good or bad?                                            |
| Preadolescent (ages 6–12)              | Industry versus inferiority     | Am I successful or worthless?                                |
| Adolescent (early teens)               | Identity versus role confusion  | Who am I?                                                    |
| Young adult (late teens and early 20s) | Intimacy versus isolation       | Shall I share my life with another person or live alone?     |
| Middle adult (late 20s to retirement)  | Generativity versus stagnation  | Will I succeed in my life, both as a parent and as a worker? |
| Older adult (after retirement)         | Ego integrity versus despair    | Have I lived a full life or have I failed?                   |



**FIGURE 10.26** In Harlow's studies, monkeys who got milk from the wire mother still clung to the cloth mother as much as they could.

sense of "trust versus mistrust." That attachment or trust comes only partly from the satisfaction of biological needs. It also depends on the emotional responses provoked by such acts as hugging.

Some highly influential evidence comes from an experiment that Harry Harlow conducted with monkeys. Harlow (1958) separated eight newborn rhesus monkeys from their mothers and isolated each of them in a room containing two artificial mothers. Four of the infant monkeys had a mother made out of wire and equipped with a milk bottle in the breast position and a mother made out of cloth with no bottle (Figure 10.26); the other four monkeys had a cloth

mother with a bottle and a wire mother with no bottle. Harlow wanted to find out how much time the baby monkeys would spend with the artificial mother that fed them.

Figure 10.27 shows the mean number of hours per day that the monkeys spent with the two kinds of mothers. Regardless of whether they got their milk from the cloth mother or from the wire mother, they all spent more than half of their time clinging to the cloth mother and very little time with the wire mother. Evidently, their attachment depended more on *contact comfort*—comfortable skin sensations—than on the satisfaction of their hunger or sucking needs.

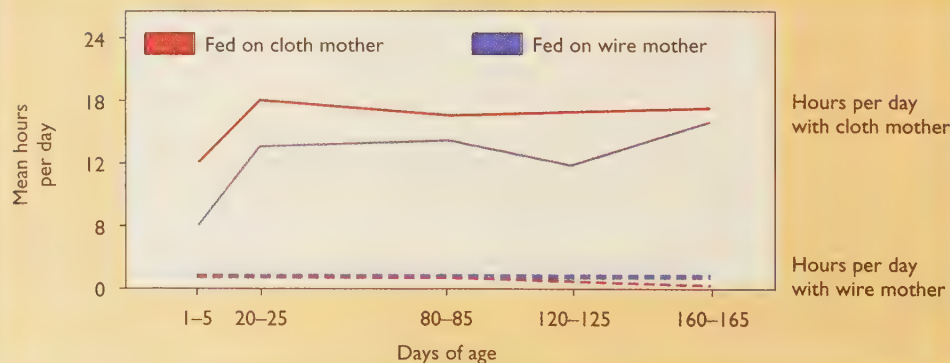
At first, Harlow thought that the cloth mothers were adequately serving the infants' emotional needs. He discovered, however, that the monkeys failed to develop normal social and sexual behavior (Harlow, Harlow, & Suomi, 1971). (See Figure 10.28.) When some of the females finally became pregnant, they proved to be woefully inadequate mothers, rejecting every attempt their babies made to cling to them or to be nursed. Clearly, the monkeys that had been reared by artificial mothers did not know how to react to other monkeys. With monkeys, as with humans, a good parent interacts with the infant, giving out social signals as well as responding to the infant's signals. Being warm and cuddly is a good start, but it is hardly enough.

What are the messages of this study? The data do *not* mean that an infant needs a mother's constant attention. They do mean that an infant needs social attention from *someone*. In later studies, Harry Harlow and Margaret Harlow (1965) found that infant monkeys reared by artificial mothers could develop fairly normally if they had frequent opportunities to play with other infant monkeys.

### Early Attachment in Humans

In the first year of life, an infant forms attachments to the parents and to many others. Suppose we want to answer some questions about early attachment: Why do some infants form stronger attachments to their parents than others do? In the long run, does a strong attachment to one person, such as the mother, produce a better psychological outcome

**FIGURE 10.27** All the baby monkeys preferred the cloth mothers, regardless of which artificial mother fed them; the two bottom lines show hours per day spent with the wire mothers.







**FIGURE 10.28** Much of what we need and learn depends on interaction with others. This mother monkey reared in isolation ignores her baby.

than moderate attachments to several people? Does a good relationship with the mother (or mother substitute) early in life make it easier for the infant to develop close bonds with other people later in life?

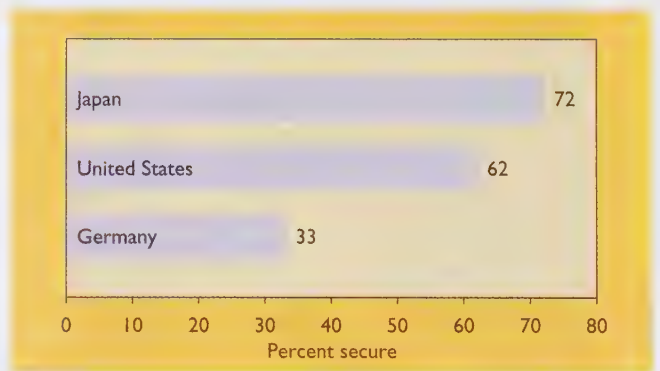
Before we can begin to answer questions like these, we first need a good way to measure attachment. One method that has proven to be both versatile and reliable is the **Strange Situation** (usually capitalized), pioneered by Mary Ainsworth (1979). In this procedure, a mother and the infant (typically 12 or 18 months old) come into a room with many toys. Then, a stranger enters the room. The mother leaves and then returns. A few minutes later both the stranger and the mother leave; then the stranger returns, and finally the mother returns. Through a one-way mirror, a psychologist observes the infant's reactions to each of these events.

Infants' responses fall into four major categories:

- **Securely attached.** The infant uses the mother as a base of exploration, often showing her a toy, cooing at her, or making eye contact with her. The infant shows some distress when the mother leaves, but cries only a little, if at all. When she returns, the infant goes to her with apparent delight, cuddles for a while, and then returns to the toys.
- **Anxiously attached.** Responses toward the mother fluctuate between happy and angry. The infant clings to the mother and cries profusely when she leaves, as if worried that the mother might not return. When she does return, the infant clings to her again.
- **Anxious and avoidant.** While the mother is present, the infant does not interact with her much and does not stay near her. The infant cries when she leaves but does not go to her when she returns.
- **Disorganized.** The infant pays little attention to the mother and seems not to notice when she leaves or returns.

Most of the existing research concerns mothers, although it has not been limited to them. As you might expect, many infants show a secure attachment not only to the mother, but also to the father (Belsky, 1996), the grandparents and other caregivers. As a rule, the quality of one relationship correlates with the quality of another; for example, if the parents have a stable marriage, usually they also develop a good relationship with their children, but if the parents have a troubled marriage, they are likely to have trouble with their children as well (Erel & Burman, 1995).

Behavior in the Strange Situation correlates strongly with behavior at home, and one can predict the pattern of attachment from how the parents treat the infant at an earlier age. The more time that parents spend affectionately interacting with an infant at the age of 3 months, the more likely that infant is to show a secure attachment at 12 months (Cox, Owen, Henderson, & Margand, 1992), and when parents learn techniques to increase their sensitivity to their infant's needs, the infant often develops a more secure relationship (van IJzendoorn, Juffer, & Duyvesteyn, 1995). Quality of attachment also depends on temperament; some infants seem to be calmer and easier to please from the start. Through a combination of parenting styles and infant temperaments, infants in Japan are more likely than U.S. infants to develop a secure attachment, and infants in northern Germany are less likely (Grossman, Grossman, Spangler, Suess, & Unzner, 1985; Miyake, Chen, & Campos, 1985):



Furthermore, an infant who shows a secure attachment to the mother is also likely to relate well to playmates (Elicker, Englund, & Sroufe, 1992). It is tempting to suggest that a secure attachment to the mother forms a model that facilitates other social relationships, but the results do not justify cause-and-effect conclusions. (For example, perhaps some infants are born with an easygoing disposition, and they therefore bring out the best in both their mothers and their playmates.) Still, we do see a consistency across relationships: Children with one good relationship are likely to have others.

An infant with an anxious or avoidant attachment is not necessarily doomed to a lifetime of difficulties, but

one with a disorganized attachment may indeed be at risk. One study found a strong relationship between a disorganized attachment in infancy and deviant or highly aggressive behavior at the age of 5 years (Lyons-Ruth, Alpern, & Repacholi, 1993).

## The Effects of Different Parenting Styles

If and when you become a parent, you will rely partly on advice from other parents, books about child care, and your own memories of how your parents treated you. Your children are excellent teachers themselves. When you do your parenting job right, they will let you know; when you make a mistake, they will definitely let you know. Parenting is difficult, but be reassured: You don't need to be perfect. Parental behavior can vary within a fairly wide range without having drastic effects on children's personality development (Harris, 1995).

Diana Baumrind (1971) conducted extensive studies of the behaviors of parents and children, identifying the various styles of parenting. The following are the most common or most basic:

**Authoritative parents:** These parents set high standards and impose controls, but they are also warm and responsive to the child's communications. They encourage their children to set their own goals and to strive toward them. Therefore, although these par-

ents set limits, they adjust those limits appropriately as the child grows older. Their children show self-reliance and self-control, cooperating with others and generally doing well in school.

**Authoritarian parents:** Like the *authoritative* parents, *authoritarian* parents set firm controls. However, they tend to be less warm and emotionally more distant from the child; they set rules without explaining why they are good rules. Their children tend to be socially responsible, law-abiding citizens, but they tend also to be discontent, distrustful, and not very independent.

**Permissive parents:** Permissive parents are warm and loving but undemanding. Their children generally lack self-control and lack a sense of social responsibility.

**Indifferent or uninvolved parents:** Some parents spend little time with their children and do little more than provide them with food and shelter. Their children, as you might imagine, tend to be impulsive and undisciplined.

We should not assume that parents' behavior causes their children's behavior. These are correlational studies, and as with nearly all correlational studies, we can imagine other possible explanations for the results. For example, children's behavior can affect their parents as much as the parents affect the children. (Perhaps children who act in self-reliant, self-controlled ways elicit authoritative parenting styles.) Another possibility is that parents and children show certain patterns of behavior because they share certain genes. (Perhaps the genes that promote permissive behavior in parents increase the probability of irresponsible behavior in children.)

Baumrind's research focused mainly on European-American children; later research found that the effects of different parenting styles vary somewhat among different ethnic groups (Darling & Steinberg, 1993). For example, the *authoritative* parenting style is associated with strong academic success and social cooperativeness for European-American children; the effects are less consistent for Asian-American and African-American families. Also, *authoritarian* European-American parents tend to have somewhat withdrawn children; authoritarian African-American parents have rather assertive children. At present, the reasons for these apparent differences are uncertain. These results remind us not to assume that the results found in one group will apply to all groups; they also emphasize the need for further research in this area.



Physical contact and cuddling are essential for the attachment that develops between parent and child. The early parent-child relationship serves as a prototype for later social attachments.

## Social Development in Childhood

The social and emotional development of children depends in part on how successful they are in forming friendships with other children (Figure 10.29). Some





**FIGURE 10.29** Children learn social skills by interacting with brothers, sisters, and friends close to their own age.

children are “popular,” having many friends and admirers. Others are “rejected,” with most other children avoiding their company. Still others are “controversial,” liked by some and rejected by others. Controversial children are generally those with some social skills but an aggressive streak. A child’s status as popular, rejected, or controversial tends to be fairly consistent from year to year (Coie & Dodge, 1983). Children with few friends tend to suffer from low self-esteem and do poorly in both schoolwork and athletics. (But does the lack of friends lead to poor performance, or does poor performance lead to a lack of friends?)

Many studies have been made of the effects of being a firstborn child or being born later. A number of those birth-order studies report that firstborn children do better in school, are more ambitious, are more honest, and have a greater need to affiliate with others. Children born later tend to be more popular, more independent, less conforming, better adjusted emotionally, and possibly more creative. These tendencies are small and inconsistent, however, and some of the evidence is based on poorly conducted studies (Ernst & Angst, 1983; Schooler, 1972). Be skeptical of recommendations that parents should space their children many years apart to give each of them the alleged benefits of the firstborn.

### SOMETHING TO THINK ABOUT

Psychologists have offered two explanations for the effects of birth order on behavior: (1) Depending on whether there are older, younger, or no other children in the family, each child is subjected to different social influences. (2) Because the mother undergoes physical changes, such as changes in her hormones after giving birth, younger children experience different prenatal influences from those experienced by the firstborn child. What kind of evidence would you need to decide whether one of these explanations was more satisfactory than the other? \*

## Adolescence

Adolescence begins when the body reaches puberty, the onset of sexual maturation. In North America, the mean ages are around 12 to 13 in girls and about a year or two later in boys. The end of adolescence is harder to identify. Adolescence merges into adulthood, and adulthood is more a state of mind than a condition of the body. Some 12-year-olds act like adults, and some 30-year-olds act like adolescents.

Adolescence is a time of transition from childhood to adulthood. Children think of themselves as part of their parents’ family; young adults are ready to become parents. Adolescents are somewhere in between, still closely tied to their parents but spending more and more time with their peer group. They do not have as much freedom and independence as adults, but they have far more than they did a few years before. Adolescence has sometimes been portrayed as a period of “storm and stress,” and for some it is indeed. Some teenagers find themselves in serious conflict with their parents (Paikoff & Brooks-Gunn, 1991); some are deeply depressed or violent; and still others make choices that will haunt them for years. Most teenagers, however, like and respect their parents and are, in turn, liked and respected by their parents. Not everyone will look back on the high-school years as “the best time of my life,” but most have a reasonably good experience of adolescence.

### Identity Development

Adolescence is a time of “finding yourself,” of determining “Who am I?” or “Who will I be?” In some societies, children are expected eventually to enter the same occupation as their parents and to live in the same town. The parents may even choose marriage partners for their children. In such societies, adolescents have few major choices to make.

Western society offers young people a great many choices. They decide how much education to get, what job to seek, and where to live, whether to marry and whom and when, what political and religious affiliation to adopt, and what standards to follow regarding sex, alcohol, and drugs. Remember from Chapter 8 that, even when people are given all the information they need to answer a question, they sometimes still rely on inappropriate heuristics and make illogical decisions. With regard to one’s most important choices, such as a career and a life partner, we must often make decisions based on very limited information. No one needs to make a perfect decision, but even making a satisfactory decision is difficult.

An adolescent’s concerns with decisions about the future and the quest for self-understanding has been called an identity crisis. The term *crisis* implies emotional turbulence, but in fact most adolescents go through this experience quite calmly. There are two major

elements of identity development—whether one is actively exploring the issue and whether one has made any decisions (Marcia, 1980). We can diagram the possibilities, using the following grid:

|                        | Has explored or is exploring the issues | Has not explored the issues |
|------------------------|-----------------------------------------|-----------------------------|
| Decisions already made | Identity achievement                    | Identity foreclosure        |
| Decisions not yet made | Identity moratorium                     | Identity diffusion          |

*Those who have not yet given any serious thought to making any decisions and who, in fact, have no clear sense of identity* are said to have **identity diffusion**. People in this stage are not actively concerned with their identity and are waiting until later to clarify the issues.

People in **identity moratorium** are *seriously considering the issues but have not yet made any decisions*. They experiment with various possibilities and imagine themselves in different roles before deciding which one is best.

**Identity foreclosure** is a state of *having firm decisions without having given them much thought*. Ordinarily, those decisions originate from one's parents or from society. For example, a young man might be told that he is expected to go into the family business with his father, or a young woman might be told that she is expected to marry and raise children. Decrees of that sort were once common in North America and Europe, and they are still common in many other societies today. Someone who accepts such decisions has little reason to explore the possibility of alternative identities.

Finally, **identity achievement** is *the outcome of having explored various possible identities and then making one's own decisions*. Identity achievement does not come all at once. For example, someone might be fully decided about a career but not about marriage, or about marriage but not a career and might be just starting to struggle with questions of religion and politics. It is also possible, even common, to reach identity achievement and then, years later, rethink some of those decisions.

## The “Personal Fable” of Teenagers

### TRY IT YOURSELF

Are the following statements true or false?

- Other people may fail to realize their life ambitions, but I will realize mine.
- I understand love and sex in a way that my parents never did.

- Tragedy may strike other people, but it will probably never strike me.
- Almost everyone notices how I look and how I dress.

According to David Elkind (1984), teenagers are particularly likely to harbor such beliefs. Taken together, he calls them the “personal fable,” the conviction that “I am special—what is true for everyone else is not true for me.” Up to a point, this fable can help us to maintain a cheerful, optimistic outlook on life, but it becomes dangerous when it leads people to take foolish chances (See Figure 10.30.)

For example, one study found that high-school girls who were having sexual intercourse without contraception estimated that they had only a small chance of becoming pregnant through unprotected sex. Girls who were either having no sex or having sex using contraception estimated a much higher probability that unprotected sex would lead to pregnancy (Arnett, 1990). Because this was a correlational study, we do not know which came first—the girls’ underestimation of the risk of pregnancy, or their willingness to have unprotected sex. Nevertheless, in either case, these results illustrate the “it can’t happen to me” attitude.

That attitude is hardly unique to teenagers, however. Middle-aged adults also overestimate their own chances of winning a lottery, regard themselves as more likely than other people to succeed on the job and as less likely than average to have an injury or a serious illness (Quadrel, Fischhoff, & Davis, 1993). That is, few people fully outgrow the “personal fable.”



**FIGURE 10.30** Some teenagers seriously risk their health and safety. According to David Elkind, one reason for such risky behavior is the “personal fable,” the secret belief that “nothing bad can happen to me.”



## Adulthood and Aging

*Adults hope that life begins at 40—but the great anxiety is that it ends there.*

—DANIEL LEVINSON (1978)

From their 20s until retirement, the main concern of most adults is: “What will I achieve and contribute to society and my family? Will I be successful at the activities I have chosen?”

The adult years begin with a flurry of major decisions about marriage, career, having children, where to live, and how to live. After one settles those decisions, adulthood becomes a time of relative security and stability. A typical middle-aged adult spends little time worrying about “what will I be doing next year?” or “who will go with me to the show on Friday night?” Whether middle age is a time of satisfaction and contentment or feeling stuck in a rut depends on the quality of the decisions made in adolescence and early adulthood.

### Job Satisfaction

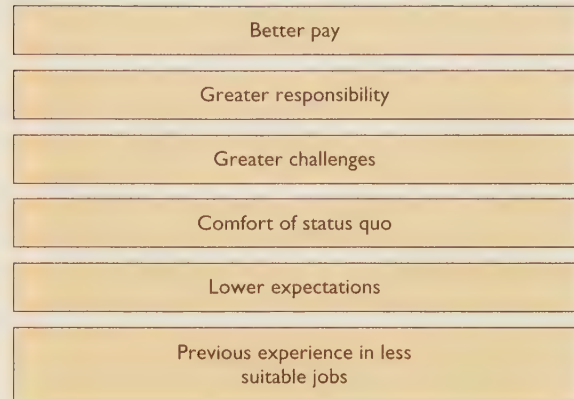
Adults who are satisfied with their job are generally satisfied with their life, and people who like their life generally like their job (Keon & McDonald, 1982). Some adults manage to be happy even though they work at an unrewarding job, but the daily work routine is bound to influence their satisfaction with life.

How satisfied *are* most workers? The answer depends on how the question is phrased. When pollsters simply ask, “Are you satisfied with your job?” about 85–90% say yes (Weaver, 1980). But when pollsters ask, “If you could start over, would you seek the same job you have now?” less than half of white-collar workers and only one fourth of blue-collar workers say yes. Although most workers say they are “satisfied” with their job, they could easily imagine being *more* satisfied.

The level of satisfaction is lower, on the average, among young workers than among older workers (Bass & Ryterband, 1979). (See Figure 10.31.) One possible explanation is that older workers have better, higher-paying jobs. Another is that today’s young people are harder to satisfy. Another possibility is that many young workers start out in the wrong job and find a more suitable one later on. Still another is that many young people are still considering the possibility of changing jobs; by age 40, most people have reconciled themselves to whatever job they have.

Your choice of career has a profound effect on the quality of your life. A student once told me that he found the courses in his major boring, but at least they were preparing him for a job. I cautioned him that he would probably find the job just as boring. Between the ages of 20 and 70, you will probably spend about half of your waking hours on the job—a long time to live with work you find unsatisfying.

Possible reasons why older workers report greater job satisfaction than younger workers



**FIGURE 10.31** Most older workers report higher job satisfaction than younger workers do. Psychologists propose several reasons for this tendency.

### The Midlife Transition

People enter adulthood with a great many hopes and goals. Then, as they settle into the daily round of activities, they tend to postpone their ambitions. Around the age of 40, some adults experience a **midlife transition, a reassessment of personal goals**. Up to this point, they had clung to the personal fable that their life would be a success in every way. There was always plenty of time to get that better job, start that family, write that great novel, take that trip up the Amazon, or get an advanced degree. But at some point, they begin to realize that the opportunity to make major life changes is rapidly fading.

The Russian novelist Leo Tolstoy wrote, at age 47:

I have lived through the period of childhood, adolescence and youth when I climbed higher and higher up the mysterious hill of life, hoping to find at its summit a result worthy of the effort put in; I have also lived through the period of maturity during which, having reached the summit, I went on calmly and contentedly . . . searching all round me for the fruits of life which I had attained . . . and I have lived through the conviction that nothing of what I expected on this summit was there and that now, only one thing remained to me—to descend to the other side to the place where I came from. And I have begun that descent. . . . I call such a condition old age, and at the present moment I have reached that condition. [Tolstoy, 1875/1978, pp. 288–289.]

Daniel Levinson (1977, 1978) reports that about 80% of all adults experience such a midlife transition. Other psychologists deny that the prevalence of the experience is anywhere near that figure. But it may only be a matter of degree: Most middle-aged adults do not experience the distress that Tolstoy described, but many go through a minor,

nontraumatic readjustment in which they review the direction that their life has been taking. When you are young, you have many goals and dreams of all the great things you will experience and accomplish. You know you are not achieving all of them, but you can tell yourself, "I'll do that later." As you get to middle age, you discover that you are running out of "later." Some of your earlier dreams are becoming unrealistic, but others are still achievable. But even if they are still achievable, you realize that time is passing: "What am I waiting for?"

The resolution of this midlife transition can take any of several directions. Some people accept their lives more or less as it is, abandon their unrealistic goals, and reset new goals that are consistent with the direction their lives have taken. Others decide that they had some earlier dreams that have not been achieved but that they are not willing to abandon. They quit their unsatisfactory jobs and go back to school, set up a business of their own, or do something else they had always wanted to try. The least satisfactory outcome is to decide, "I can't abandon my dreams, but I can't do anything about them either. I need to keep this job that I hate and press on with a life that has no joy."

The moral is clear: To increase your chances of feeling good in middle age and old age, make the right decisions now. If there is something you really care about, something you will feel bad about if you don't at least try it, get started on it now. Take chances on yourself. If you fail, well, at least you tried, and you won't always wonder what would have happened if you had only tried. Besides, in the process of trying to do what you want, you might discover a related opportunity that you would not have found otherwise. And you never know: You just might succeed.

## Old Age

The percentage of people who live into their 70s and 80s has risen steadily throughout the 20th century. Their health has also improved, and so has their activity level and intellectual performance (Schaie, 1994). People age in different ways. Some people, especially those with Alzheimer's disease or other serious ailments, deteriorate both intellectually and physically—in some cases, quite rapidly. Other older people remain almost as active and alert as ever, well into their 80s or even 90s.

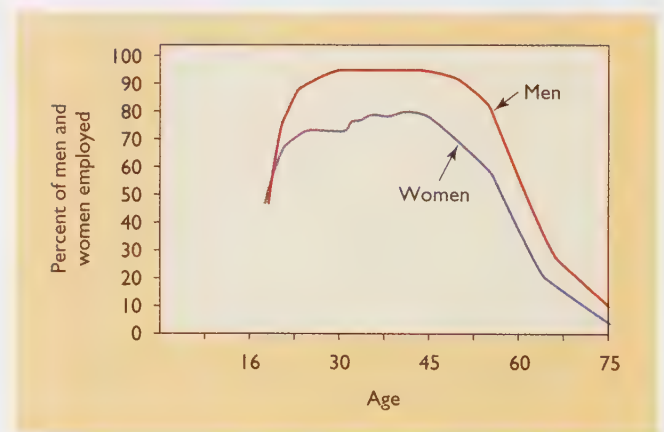
One common concern of old age is to maintain a sense of dignity and self-esteem. How well older people maintain their dignity depends largely on how they are treated by their family, their community, and their society (Figure 10.32). Some cultures, including the people of Korea, observe a special ceremony to celebrate a person's retirement or 70th birthday (Damron-Rodriguez, 1991). African-American and Native American families traditionally honor their elders, giving them a position of status in the family and calling upon them for advice. Japanese families follow a



**FIGURE 10.32** In Tibet and many other cultures, children are taught to treat old people with respect and honor. Most older people say that maintaining a sense of dignity is very important to them.

similar tradition, at least publicly (Koyano, 1991). Dignity and self-esteem also depend upon a feeling of having lived life well. To feel dignity in old age, spend your time well in your youth.

As people age, they have to deal with changes in social status as well as changes in their bodies. Figure 10.33 shows the percentage of people in the United States who



**FIGURE 10.33** The percentage of people employed rises and falls as a function of age. These figures include students, patients in hospitals, and others who are not looking for a job. (Based on data from U.S. Department of Labor, 1989.)



are employed at various ages (U.S. Department of Labor, 1989). For about 40 to 50 years of life, people spend much of their time at work, taking pride and gaining prestige from their accomplishments. Then sometime between the ages of 60 and 70, they face voluntary or mandatory retirement. Many continue to be active, doing volunteer work, taking a part-time job . . . or serving as a U.S. senator. . . .

People adjust to retirement in different ways (Atchley, 1980). Those who were active and had a variety of interests before retirement usually find the adjustment easiest. Most retirees go through a “honeymoon” period at first, doing all the things they never had time to do before. However, their reborn hopes for great achievement soon fade. Just as many 20-year-olds start out with lofty goals that they can never achieve, many 65-year-olds enter retirement with unrealistic expectations. Before long, they review their prospects and revise their goals.

It is important for old people to maintain some sense of control over their lives, even if their health fails. Consider someone who has spent half a century managing a household or running a business, who is now living in a nursing home where staff members make all the decisions, from scheduling meals to choosing television programs. This loss of control can be both frustrating and degrading. When the nursing home staff leaves some of the choices up to the residents and lets them perform some tasks by themselves, their health, alertness, and memory tend to improve (Rodin, 1986; Rowe & Kahn, 1987).

## The Psychology of Facing Death

*A man who has not found something he is willing to die for is not fit to live.*

—MARTIN LUTHER KING, JR. (1964)

*This is perhaps the greatest lesson we learned from our patients: LIVE, so you do not have to look back and say, “God, how I have wasted my life!”*

—ELISABETH KÜBLER-ROSS (1975)

*The worst thing about death is the fact that when a man is dead it's impossible any longer to undo the harm you have done him, or to do the good you haven't done him. They say: live in such a way as to be always ready to die. I would say: live in such a way that anyone can die without you having anything to regret.*

—LEO TOLSTOY (1865/1978, p. 192)

We commonly associate death with older people, although a person can certainly die at any age. Not only do we have trouble dealing with the prospect of our own death, but we also find it difficult to deal with the death of others. Each society provides standards and guidance for how people should deal with the death of a loved one, and Western society has changed its standards over the last century. In the 1800s, the death of a spouse was expected to lead to a



Linda McCartney battled cancer for several years before her death in 1998. In the “acceptance” stage of facing death, people take a realistic approach to their impending death.

long-lasting, almost permanent grief. Today, society encourages people to try to return to normal functioning (Stroebe, Gergen, Gergen, & Stroebe, 1992). Most people need about a year to work through their grief, but the course of bereavement varies greatly from one person to another.

Most people deal with anxiety about their own death by telling themselves that death is far in the future. When they learn that they have a fatal disease, they react in many ways. Elisabeth Kübler-Ross (1969), in her book *On Death and Dying*, suggests that dying people go through five stages of adjustment, as listed below. Dying people can move from stage to stage in any order and sometimes skip some stages altogether.

- **Denial:** the refusal to accept the information about having an incurable condition. The person visits doctors, faith healers, and anyone else who promises a chance for recovery.
- **Anger:** rage at doctors, nurses, and relatives. One characteristic utterance is, “Why me?”
- **Bargaining:** promising good behavior in the future in return for recovery. Religious people often make promises to God; some people also try offering the doctor a special inducement.
- **Depression:** sadness about losing life. Some people become very depressed, but not all do.
- **Acceptance:** a readiness to die. People who reach this stage lose interest in most of what is happening around them.

## SOMETHING TO THINK ABOUT

Do Kübler-Ross’s stages apply only to people who are dying? Or do people react in similar fashion to lesser losses, such as a poor grade or being fired from a job? Have you ever had a personal experience when you went through some of the same stages? ★

## THE MESSAGE

### *Social and Emotional Issues Through the Life Span*

People confront crises and issues at different ages, but whatever one does at one age strongly affects life at a subsequent age. A young child who fails to develop a sense of trust or autonomy, or an adolescent who fails to achieve a sense of identity, enters the next age with some catching up to do. A young adult who makes poor decisions about marriage and career—sometimes no decision at all is the worst decision—causes problems that may linger into middle age and beyond. People do recover from earlier problems: The point is simply that issues that we face at one age may persist to later stages of life. No stage of development is ever completely over.

## SUMMARY

- ★ **Cross-sectional and longitudinal studies.** Psychologists study development by means of cross-sectional studies, which examine people of different ages at the same time, and by means of longitudinal studies, which look at a single group of people at various times as they grow older. Each method has its advantages and disadvantages. (page 368.)
- ★ **Cohort effects.** In some cases, a difference between young people and old people is not due to age itself but to a cohort effect: People born in one era differ from those born in a different era. (page 369)
- ★ **Erikson's ages of development.** Erik Erikson described the human life span as a series of eight ages, each with its own social and emotional conflicts. (page 370)
- ★ **Infant attachment.** The initial attachment of an infant to his or her mother depends on the comfort of physical contact rather than on being fed. Later, other kinds of parent-child contact become increasingly important for developing the infant's social behaviors. (page 371)
- ★ **Parenting styles.** Different parenting styles are associated with varying behavior patterns in children; the nature of that association appears to vary across ethnic groups. (page 374)
- ★ **Social development of children.** The social development of a child depends on the influences of other children, including brothers and sisters. (page 374)
- ★ **Adolescent identity crisis.** Adolescents have to deal with an identity crisis, the question "Who am I?" Many experiment with several identities before deciding which one seems right. (page 375)
- ★ **Adults' concerns.** One of the main concerns of adults is balancing the competing demands of family and career. For most adults, satisfaction with life is closely linked with their satisfaction on the job. Some adults experience a midlife transition and reevaluate their goals. (page 377)
- ★ **Old age.** In old age, people make new adjustments, including the adjustment to retirement. Maintaining dignity and independence is a key concern. (page 378)

- ★ **Facing death.** People at all ages must face the anxieties associated with the inevitability of death. People go through some characteristic reactions when they know that they are likely to die soon. (page 379)

## Suggestions for Further Reading

- Elkind, D. (1984). *All grown up and no place to go*. Reading, MA: Addison-Wesley. An account of the problems that teenagers and young adults face.
- Kail, R. V., & Cavanaugh, J. C. (1996). *Human development*. Pacific Grove, CA: Brooks/Cole. A good textbook on all aspects of developmental psychology.

## Terms

- cross-sectional study** a study of groups of individuals of different ages all at the same time (page 368)
- longitudinal study** a study of a single group of individuals over time (page 368)
- selective attrition** the tendency of some kinds of people to be more likely than others to drop out of a study (page 369)
- cohort** a group of people born at a particular time (as compared to people born at different times) (page 369)
- basic trust versus mistrust** the conflict between trusting and mistrusting that parents and other key figures will meet one's basic needs; the first conflict in Erikson's eight ages of human development (page 370)
- autonomy versus shame and doubt** the conflict between independence and doubt about one's abilities (page 370)
- initiative versus guilt** the conflict between independent behavior and behavior inhibited by guilt (page 370)
- industry versus inferiority** the conflict between feelings of accomplishment and feelings of worthlessness (page 370)
- identity versus role confusion** the conflict of discovering who one is or will be (page 371)
- intimacy versus isolation** the conflict between spending one's life with a partner and living alone (page 371)
- generativity versus stagnation** the conflict between a productive life and an unproductive one (page 371)
- ego integrity versus despair** the conflict between satisfaction and dissatisfaction with one's life; the final conflict in Erikson's eight ages of human development (page 371)
- attachment** a long-term feeling of closeness between people, such as a child and a caregiver (page 371)
- Strange Situation** procedure in which a psychologist observes an infant's behavior in an unfamiliar room when the mother leaves and returns (page 373)
- authoritative parents** those who are demanding and impose firm controls, but who are also warm and responsive to the child's communications (page 374)
- authoritarian parents** those who exert firm controls on their children, generally without explaining the reasons for the rules and without providing much warmth (page 374)
- permissive parents** those who are warm and loving but undemanding (page 374)
- indifferent or uninvolved parents** those who pay little attention to their children beyond doing what is necessary to feed and shelter them (page 374)
- identity crisis** concerns with decisions about the future and the quest for self-understanding (page 375)



- identity diffusion** the condition of having not yet given any serious thought to identity decisions and having no clear sense of identity (page 376)
- identity moratorium** the state of seriously considering one's identity without yet having made any decisions (page 376)
- identity foreclosure** the state of having made firm identity decisions without having thought much about them (page 376)
- identity achievement** the outcome of having explored various possible identities and then making one's own decisions (page 376)
- midlife transition** a time of goal reassessment (page 377)

## Answers to Concept Checks

- Use a longitudinal study. A longitudinal study studies the same people repeatedly instead of comparing one cohort with another. (page 369)
- Use a cross-sectional study, comparing people of different ages all in the same year. (page 370)
- Another possible explanation is that the first-year students who have the lowest grades (and therefore pull down the grade average for first-year students) do not stay in school long enough to become seniors. (page 370)

## Web Resources

### The Child Psychologist

[www.childpsychology.com/](http://www.childpsychology.com/)

Rene Thomas Folse's site focuses on "Specific Disorders and Other Reasons for Concern" about the behavior of children, and on "Treatment, Resources and Remediation." There are links to information about several specific diagnostic tests used with children.

# Growing Up the Same and Different: Temperament, Family, Gender, and Cultural Influences

Do people's temperamental differences remain consistent as they grow older?

How does family life guide children's social and emotional development?

How do boys and girls differ, and why?

How does ethnic identity influence development?

If you had to describe yourself in 200 words or less, what would you say? You would probably mention your age, whether you are male or female, where you live, your main interests and activities, and something about your family. You might mention your political or religious affiliation, if these are important to you. Would you also mention your ethnic identity? Some would; some would not.

The point is this: You are a unique individual, and you are also a member of various identifiable groups. Your group memberships mold what you think of yourself and how other people treat you. You are a complex product of both what you bring to a situation (your personality and temperament) and how a situation affects you. Let's begin with temperament.

## Temperament and Lifelong Development

People differ markedly in their temperament—their tendency to be either active or inactive, outgoing or reserved. Would you rather go to a party where you will meet new

people, or would you prefer to spend a quiet evening with a few old friends? Do you like to try a new, somewhat risky adventure, or would you prefer to watch while someone else tries it first? In general, are you more impulsive or more reserved than most of the people you know?

Now, consider how you just described yourself: Is that the way you have always been, more or less? Or have you changed considerably? Were you at one time a great deal more outgoing and adventurous, or more shy and reserved, than you are now?

According to the research, most people are fairly consistent in their temperament over long periods of time. We can begin to measure temperament even before birth. During the last two months before birth, infants differ substantially in the amount of kicking and other movements, and the ones that are most active before birth tend to remain the most active after birth (Eaton & Saudino, 1992).

Jerome Kagan and Nancy Snidman (1991) measured how often 4-month-old infants kicked, how often they cried, and how tense their hands were. A few months later, they examined the same infants' responses to mildly frightening situations. (For example, the experimenter might uncover a rotating toy, frown, and scream a nonsense phrase.) Infants who showed the most kicking, crying, and tension at the age of 4 months also tended to show the most fears at the ages of 9 and 14 months. That is, their temperaments were consistent from one test to the next.

## CONCEPT CHECK

8. Was Kagan and Snidman's study longitudinal or cross-sectional? (Check your answer on page 388.)

Infants who seldom kick, cry, or display fear are called "easy" or "uninhibited" (Thomas & Chess, 1980; Thomas, Chess, & Birch, 1968). Easy infants develop regular sleeping and eating habits, show interest in new people and new objects, and are easily comforted. The kicking, crying, highly fearful infants are, as you might guess, termed "difficult" or "inhibited." Their eating and sleeping habits are irregular, they show frequent signs of tension, and they are hard to comfort (Kagan, 1989). These infants are also more likely to contract various contagious diseases (Lewis, Thomas, & Wrobley, 1990). Evidently, temperament is connected to all aspects of how the body functions. Not all infants fit into either the "inhibited" or "uninhibited" categories; many are

Most people's temperaments remain fairly consistent over time. An outgoing, uninhibited child will probably continue to be outgoing and uninhibited for years, perhaps even into adulthood.





intermediate. Some fit a special category called “slow to warm up”: They withdraw at first from unfamiliar people and new experiences, but after repeated exposures they begin to react positively.

How long do infant temperaments last? The answer varies. Infants identified as highly inhibited at the age of 21 months are more likely than others to end up as shy, quiet, nervous, and fearful 7½-year-olds (Kagan, Reznick, & Snidman, 1988). Similarly, many uninhibited infants develop into socially interactive, highly talkative 7½-year-olds. However, major changes occur, too; about 10% of uninhibited infants develop into shy 7½-year-olds (Kagan, 1989).

What causes differences in temperament? Genetic differences make some contributions. Monozygotic (identical) twins resemble each other in temperament more than dizygotic (fraternal) twins do (Matheny, 1989). Monozygotic twins reared in separate and apparently rather different environments still generally end up with similar temperaments (Bouchard, Lykken, McGue, Segal, & Tellegen, 1990). Environmental factors obviously play a major role also; otherwise, monozygotic twins would always exactly match each other in temperament.

Heredity and environment can interact in some complex ways to influence temperament. Furthermore, a child's developing temperament can alter the environment, which in turn influences the further development of temperament (Bouchard et al., 1990; Collins & Gunnar, 1990). For example, a child with an uninhibited temperament will meet more people and try more new experiences than an inhibited child will; while doing so, the child learns new skills and develops new behaviors that can alter later reactions to new situations.

## The Family

One of the most powerful influences in the human environment is the family. Children in a loving family gain a sense of security and learn, “I am a lovable person.” By playing with brothers, sisters, or other children, they begin to learn social skills. They also learn social skills by observation, especially by watching how their parents relate to each other. How do variations in early family environments affect a child's social and emotional development?

### Parental Employment and Child Care

Many people assume that the “normal” way of rearing infants is for the mother to stay with them full-time, because that custom was prevalent in much of North America and Europe for many years. However, child-rearing customs vary greatly from one culture to another, one era to another, even from one social class to another (see Figure 10.34). In many cultures, especially subsistence cultures, a mother returns to her usual tasks of gathering food and so forth shortly after giving birth, leaving her infant most of the day



**FIGURE 10.34** In many cultures, it has long been the custom for a mother to leave her infant for much of the day with friends, relatives, and other children. An infant with multiple caregivers can develop just as healthy emotional and social behaviors as one having a single caregiver.

with other women, relatives, and any older children in the family (McGurk, Caplan, Hennessy, & Moss, 1993). For example, in the Efe culture of Africa, a mother stays with her infant only about half of the day, but the infant is seldom alone. Within the first few months, the infant will establish strong attachments to several adults and children (Tronick, Morelli, & Ivey, 1992). In wealthy families in Europe, it has long been the custom for “nannies” (paid caregivers) to take care of the children for most of the day.

Still, many psychologists in Europe and North America developed a theoretical belief that healthy emotional development required an infant to establish a strong attachment to a single caregiver—ordinarily, the mother. When more and more families began placing infants in day care so that both parents could return to work shortly after their infant's birth, a question arose about whether those children would be psychologically harmed.

The research on day care indicates that children from good homes generally develop satisfactorily, both intellectually and socially, if they receive at least adequate day care (Scarr, 1998). The quality of the day care makes a greater difference for children from disadvantaged homes. (Evidently, development is satisfactory when either home life or day care is good; problems arise when both are bad.) When children from disadvantaged homes get good day care (with a consistent, affectionate staff and adequate facilities), they develop better than if they had stayed at home (McGurk et al., 1993; Scarr, Phillips, & McCartney,

1990). Children from deprived homes who also get poor day care—which is often all the parents can afford—are at a definite disadvantage.

The most enduring controversy has concerned the effects of starting day care before a child is 1 year old. Several studies have reported that children who attend day care for 20 hours or more per week during the first year of their life fail to develop a secure relationship with their mothers, as measured in the Strange Situation (discussed earlier). However, the problem may be in the measurement (McGurk et al., 1993). Remember, in the Strange Situation, the mother leaves the infant in a room with a stranger and later returns; an infant who does not greet her return eagerly is considered to have an avoidant relationship. But think about this situation from the standpoint of an infant who is left in day care 5 days per week: Your mother has frequently left you in a room with a stranger before. Now she has done it again. Every time she left you in the past, you have been safe and well cared for; so, it's no big deal. When she returns three minutes later . . . perhaps the main surprise is that she came back so soon.

In other words, failing to greet the mother on her return might have different implications for an infant accustomed to day care than it does for an infant accustomed to spending all day with the mother. The research to date does not document that day care harms the infant's social development, provided that the day care is of good quality. (Unfortunately, of course, affordable good-quality day care is often hard to find.)

## Nontraditional Families

The “traditional” family consists of a mother, a father, and their children. Many people who grew up in a reasonably happy traditional family think of such an arrangement as so “right” that anything else must be wrong.

Nevertheless, most children grow up in families that are significantly different from the traditional model. For example, psychologists have studied the children of unmarried mothers (Weissman, Leaf, & Bruce, 1987), children reared by a gay or lesbian couple (Patterson, 1994), and children of families where the mother works full-time and the father stays home with the children (Parke, 1995). Parenting styles do differ a bit; for example, fathers generally play with their children more and talk to them less. However, the usual finding is that children in nontraditional families turn out about the same as those from more traditional families. Many of the differences that do emerge are due to economics: Most unmarried mothers have very limited finances and cannot provide their children with the same opportunities that middle-class children have.

## Parental Conflict and Divorce

At an earlier time in the United States, divorce was unusual and frowned upon. When Adlai Stevenson was defeated in the presidential campaign of 1952, one explanation given

was that American voters would never elect a divorced candidate as president. By 1980, when Ronald Reagan was elected president, hardly anyone considered his divorce and remarriage to be an issue at all. By then, divorce was simply a fact of life.

An estimated 75% of African-American children and 38% of European-American children will experience the divorce of their parents before the children reach the age of 16. Most of those children show a variety of academic, social, and emotional problems, compared to children in two-parent households. One reason is that the children of divorced families receive less attention and suffer greater economic hardship. The main reason, however, is that children in divorced families often endure the prolonged conflict and hostility between their parents (Amato & Keith, 1991). If the divorce takes place while the children are still too young to realize what is happening, the effects on the children will be milder (Tschann, Johnston, Kline, & Wallerstein, 1990).

Mavis Hetherington and her associates have conducted longitudinal studies of middle-class, elementary-school children and their families following a divorce (Hetherington, 1989; Hetherington, Cox, & Cox, 1982). In each case, the mother had custody of the children. Hetherington found that most of the children and many of the parents suffered considerable upheaval after the divorce, especially during the first year.

During this first year, the children resorted to pouting and other forms of seeking attention. They were generally angry about their parents' divorce, and they let the parents know about it. Many of the divorced mothers had very difficult relationships with their sons; one exasperated divorced mother described their relationship as “like getting bitten to death by ducks.” Boys in particular became very aggressive toward other children, and much of their aggressive behavior was unprovoked and ineffective (Figure 10.35). After 2 years following the divorce, these boys became better adjusted, but by then they had been rejected by their peers.



**FIGURE 10.35** Sons of divorced parents often go through a period when they act out their frustrations by starting fights.



Boys who changed schools after the first year managed to escape their reputation and make a fresh start.

The degree of distress varied from one child to another. Generally, a child's distress was greater if the mother had not worked before the divorce and had taken a job immediately afterward—often an economic necessity. The children in such families felt that they had lost both parents. In the studies by Hetherington and her associates, boys showed more distress and more negative behavior than girls did, partly because the mothers retained custody of the children. When the father had custody, the boys reacted better than the girls did (Santrock, Warshak, & Elliott, 1982). (Because many other factors are at work, we cannot conclude, however, that a father should always be granted custody of sons or that a mother should always be granted custody of daughters.)

In families where the mother remarried, the children—especially the girls—were often indifferent or hostile to the stepfather and showed poorer adjustment than children of mothers who did not remarry (Hetherington, Bridges, & Insabella, 1998). Many of the girls rejected every attempt their stepfathers made to establish a positive relationship, until eventually the stepfather simply gave up (Hetherington, 1989). When stepfathers were asked to name “all the members of your family,” most did not even mention their stepdaughters.

Hetherington's studies concentrated on European-American, middle-class children, and the results are somewhat different for other cultures. Divorce is more common in African-American families, but in most cases it is more accepted and less stressful (Fine & Schwebel, 1987). Many African-American families ease the burden of single parenthood by having a grandmother or other relative share in the child care. As in European-American families, the more upset the mother is by the divorce, the more upset the children are likely to be (Phillips & Alcebo, 1986).

Exceptions can be found to almost any generalization about the effects of divorce on children, however (Hetherington, Stanley-Hagan, & Anderson, 1989). Some children show emotional distress for a year or two and then gradually feel better. Others continue to act depressed 5 or 10 years after the divorce. A few seem to do well for a while and then show signs of distress years later, especially during adolescence. Other children are amazingly resilient throughout their parents' divorce and afterward. They keep their friends; they do all right in school; they maintain good relationships with both parents. Generally, these children were well-adjusted before the divorce, and their parents displayed a minimum of conflict toward each other (Hetherington, 1989; Kline, Tschann, Johnston, & Wallerstein, 1989).

Given the emotional trauma commonly associated with divorce, should parents stay together “for the children's sake”? The answer depends on how serious the parents' conflicts are. Children who grow up in households where the parents are constantly angry or distressed develop emotional problems similar to those of the children of divorced families

(Emery, 1982). Indeed, most children (especially boys) in divorced families begin to show signs of distress years *before* the divorce itself, perhaps in response to the parental conflict they already experience (Cherlin et al., 1991).

## The Influence of Gender

How would you be different if you had been born female instead of male, or male instead of female? We all have certain impressions of how boys act differently from girls, and men from women. But how many of those impressions are correct, how many are exaggerations, and how many are simply false? Remember the phenomenon of *illusory correlations* from Chapter 2: We tend to remember most clearly the examples that fit a pattern we expect to find; therefore, we convince ourselves that our expectations were correct. Sex differences do occur, but we must base our conclusions on systematic data and not on casual impressions.

### Sex Differences in Behavior

In 1974, Eleanor Maccoby and Carol Jacklin published an extensive review of the literature on sex differences in behavior. They concluded that the evidence supported a few generalizations. For example, females tend to perform better on certain aspects of language use, whereas males tend to perform better on certain mathematical and visual-spatial tasks. These differences are marked in adolescents and adults but questionable in children. Also, males tend to fight more. Sex differences in other aspects of behavior were possible but not clear from the evidence (Maccoby & Jacklin, 1974).

Maccoby and Jacklin's review drew two kinds of criticisms. First, some argued that they had either ignored or understated many behavioral differences between the sexes: On the average, men swear more than women. Women tend to know more than men do about flowers. Men and women generally carry books and packages in different ways (Figure 10.36). The list of miscellaneous differences could go on and on.

Secondly, others argued that Maccoby and Jacklin had overstated the differences and that in fact the behavior of males and females hardly differs at all. According to this point of view, boys and girls—and men and women—act differently only because our society tells them to. (Actually, even if this idea is true, it is not a fair criticism of Maccoby and Jacklin, who were trying to describe the differences, not explain them.)

Much time has passed since Maccoby and Jacklin's 1974 review. Most of the same conclusions still hold, although we must modify them somewhat (Maccoby, 1990). For example, the female advantage on verbal tasks seems to have faded (for unknown reasons). Also, it now appears that women are, on the average, more easily influenced by other people's opinions. (Depending on your point of view, you